

# MPEG-2 Recording Software Usage Instructions

- Windows 2000/XP
- Linux

## 1. Introduction

The DEKTEC DtRecord software is an elementary command-line program to record an MPEG-2 Transport Stream into a file with the DTA-120 (DVB/ASI input), the DTA-122 (DVB/SPI input) or the DTA-140 (DVB/ASI input+output) PCI card.

The package includes the DtRecord source code, to (1) illustrate the use of the DTAPI library for building digital-video applications and (2) to provide a basis for creating a custom Transport-Stream recorder application.

The DtRecord package consists of the following files:

| File                                            | Version                           | Description                                    |
|-------------------------------------------------|-----------------------------------|------------------------------------------------|
| <a href="#">DtRecord.exe</a>                    | <a href="#">none</a> <sup>1</sup> | Compiled recorder software                     |
| <a href="#">DtRecord.cpp</a>                    | <a href="#">v2.0.1</a>            | DtRecord source file with all application code |
| <a href="#">Makefile</a>                        | –                                 | Makefile for compiling DtRecord under Linux    |
| <a href="#">DtRecord Usage Instructions.pdf</a> | <a href="#">v2.0.1</a>            | This document                                  |

## 2. Running the DtRecord application

The following items are required to run the DtRecord application:

- The DtRecord executable file;
- The Dta1xx driver<sup>2</sup>, installed on the target system (the latest version of the driver can be downloaded from [www.dektec.com](http://www.dektec.com));
- One or more DTA-120, DTA-122 and/or DTA-140 PCI cards installed in the target system.

After start-up, the DtRecord application first prints the specified settings to the console<sup>3</sup>. Then recording commences. Recording stops when until the record file reaches its maximum size (as specified on the command line), the user presses a key or an error occurs. Please refer to the Troubleshooting section (§5) for help on errors.

<sup>1</sup> DtRecord.exe does not contain a version resource. "DtRecord -?" prints out the version number.

<sup>2</sup> The Linux version of the Dta1xx driver is part of the Linux SDK.

<sup>3</sup> Provided that the *run-silent* option (*-s*) has not been specified.

## 3. Command-Line Usage

### 3.1. Syntax

DtRecord **recfile**<sup>4</sup> [-x MaxSize] [-m Mode] [-t Type] [-n Number] [-s] [-?]

### 3.2. Options

- x** Maximum file size in MB. If this option is not specified, the size of the record file is unbounded
- m** Receive Mode (default **ST188**). See §3.2.1 for a more detailed description
  - Use: **ST188** Store packets as 188-byte packets
  - ST204** Store packets as 204-byte packets
  - STRAW** No notion of packets. Store all valid bytes.
- t** Board Type to use (default: any input board<sup>5</sup>)
  - Use: **120** for DTA-120
  - 122** for DTA-122
  - 140** for DTA-140
- n** Board Number to use (default **1**)
  - Note: **1** indicates the first board
- s** Run in silent mode (i.e. no messages printed to the screen)
- ?** Display command line help

#### Note:

- The first option should either be the **recfile** or the help option.

#### 3.2.1. Receive Mode

This section provides a more detailed description of the Receive-Mode option. For a complete description of the Receive Modes refer to the datasheets of the PCI input card that is used. As mentioned above the Receive-Mode option supports the following five modes:

- **ST188**: This mode can be used if the **recfile** should contain 188-byte packets only. In this mode, the DEKTEC input board will remove the trailing 16 bytes of each incoming 204-byte packet; 188-byte packets are left untouched. If the hardware is not synchronised to the incoming Transport Stream, all packets will be dropped until synchronisation is achieved.
- **ST204**: This mode can be used if the **recfile** should contain 204-byte packets only. In this mode, the DEKTEC input board will add 16 dummy bytes to each incoming 188-byte packet; 204-byte packets are left untouched. If the hardware is not synchronised to the incoming Transport Stream, all packets will be dropped until synchronisation is achieved.
- **STRAW**: In this mode, the DEKTEC input board stores *all* valid data bytes without regard to packet boundaries (i.e. no synchronisation required). This mode is especially helpful for analysing the cause of a synchronisation problem (e.g. packets do not start with a sync byte)

<sup>4</sup> Name of the file in which the Transport Stream is recorded.

<sup>5</sup> Any input board means the Nth input board (either DTA-120, DTA-122 or DTA-140) found in the system, where N is the board number specified with the **-n** option.

or invalid packet sizes).

### 3.3. Examples

```
DtRecord myfile.ts
```

```
DtRecord "c:\Video streams\myfile.ts" -x 100 -t 122 -m STRAW -n 2
```

## 4. Compiling the DtRecord Application

### 4.1. Compilation under Windows

Before compiling `DtRecord.cpp`, be sure to include the (path to the) DTAPI header file (`dtapi.h`) and to link with the library (`dtapi.lib`). The latest version of these files is available on the DEKTEC website ([www.dektec.com](http://www.dektec.com)).

The DTAPI library has been compiled with the C/C++ Code-Generation Options in VC++ set to: "Use run-time library: Multithreaded DLL." On the compiler command line this corresponds to the `/MD` option. DtRecord has to be compiled with this same setting, or a conflict with the version of the run-time library will occur.

### 4.2. Compilation under Linux

The DtRecord package comes with a makefile intended for compilation on a Linux machine. Before compiling make sure the makefile uses the correct include paths to the DTAPI include file (`dtapi.h`) and the DTAPI library file (`dtapi.o`). To compile, type 'make' in the directory containing `DtRecord.cpp`.

The DTAPI library for Linux is part of our Linux SDK, which can be downloaded from the DEKTEC website ([www.dektec.com](http://www.dektec.com)).

## 5. Troubleshooting

This section provides help for problems that might occur while using the DtRecord application. If the problem persists refer to §6.

### 5.1. DtRecord Errors

The DtRecord application can generate two types of errors, namely: command-line errors and application errors. Command-line errors are caused by missing or invalid options. Application errors indicate a problem with the system.

### 5.1.1. Command-Line Errors

The table below shows the possible command-line error messages and their probable cause(s):

| Message:                                    | Cause(s):                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "Missing or invalid command line option(s)" | <ul style="list-style-type: none"> <li>An invalid option parameter has been specified (e.g. character(s) instead of a number, negative numbers or unknown parameter);</li> <li>An option has been specified more than once;</li> <li>An option is unknown;</li> <li>The option prefix is missing (all options except the <b>recfile</b> should start with '/' or '-', followed by the option identifier and an optional parameter).</li> </ul> |
| "No record file specified"                  | <ul style="list-style-type: none"> <li>The <b>recfile</b> has not been specified as the first argument</li> </ul>                                                                                                                                                                                                                                                                                                                              |

### 5.1.2. Application Errors

Application-error messages start with "Error: ", followed by the error cause.

The table below shows the possible application error messages and their probable cause(s):

| Message:                                                                                   | Cause(s):                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "Can't set Receive Mode"                                                                   | <ul style="list-style-type: none"> <li>The specified Receive Mode is not supported by the DTA-120, DTA-122 or DTA-140</li> </ul>                                                                                                                                                             |
| "Can't open <b>recfile</b> for writing"                                                    | <ul style="list-style-type: none"> <li>The specified <b>recfile</b> is not accessible. Possibly the file is used by another application or the user does not have access rights for the file/directory.</li> </ul>                                                                           |
| "FIFO underflow. Can't write data fast enough"                                             | <ul style="list-style-type: none"> <li>The system is not fast enough for recording data at the current input rate: the hard-disk may not be fast enough or not enough memory is installed.</li> <li>Other applications are performing processor and/or hard-disk intensive tasks.</li> </ul> |
| "No input board in the system" or "No DTA-1xx in the system"                               | <ul style="list-style-type: none"> <li>No DTA-120, DTA-122 or DTA-140 board is installed in the system;</li> <li>The DTA-1xx driver is not installed or not working correctly<sup>6</sup>.</li> </ul>                                                                                        |
| "Could not find input board #X in the system" or "Could not find DTA-1xx #X in the system" | <ul style="list-style-type: none"> <li>The Xth board as specified by the Board-Number option is not present in the system;</li> <li>See also item above.</li> </ul>                                                                                                                          |

DtRecord can generate more application errors than specified above. However, the cause of such errors is not always evident and may require support from DEKTEC.

<sup>6</sup> Refer to the troubleshooting section in the Dta1xx driver installation guide for more details.

## 6. Contacting DEKTEC for Support

If the troubleshooting section does not provide a solution for your problem or the error(s) does persist you may send an email to [support@dektec.com](mailto:support@dektec.com).

Please specify the following items in your problem report:

- The error message generated by the DtRecord application;
- The type of board used (DTA-120, DTA-122 or DTA-140);
- Operating System;
- Driver Version;
- DTAPI library version (for custom applications).

## 7. DtRecord Revision History

| Revision               | Date       | Changes                                                                                                                                                                                                                                                          |
|------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">v2.0.1</a> | 2003.20.01 | <ul style="list-style-type: none"><li>• Fix recording bug for DTA-140</li><li>• Set Receive-Control field to idle when done, to avoid Receive-FIFO overflow (with corresponding green/red flashing of the status LED) some time after running DtRecord</li></ul> |
| <a href="#">v2.0.0</a> | 2002.12.06 | <ul style="list-style-type: none"><li>• Can now be compiled both under Linux and under Windows</li></ul>                                                                                                                                                         |
| <a href="#">v1.1.0</a> | 2002.11.14 | <ul style="list-style-type: none"><li>• Support for DTA-140</li></ul>                                                                                                                                                                                            |
| <a href="#">v1.0.1</a> | 2002.07.26 | <ul style="list-style-type: none"><li>• Fix bug to recognize DTA-120 without /t option</li></ul>                                                                                                                                                                 |